Putnam POTD November 19, 2025

I. PROBLEM

(2017 A6) The 30 edges of a regular icosahedron are distinguished by labeling them $1,2,\ldots,30$. How many different ways are there to paint each edge red, white, or blue such that each of the 20 triangular faces of the icosahedron has two edges of the same color and a third edge of a different color? [Note: the top matter on each exam paper included the logo of the Mathematical Association of America, which is itself an icosahedron.]

II. SOLUTION

We assign numbers $0,1,2$ to the edge colors white, red, blue, and consider the labels each face of the icosahedron under mod 3, where the label is equal to the sum of the three edges.

If the three edges have the same color x, the value of the face is $3x = 0$. 1

If the three edges all have different colors $x,y,z = 0,1,2$, the value of the face is $x+y+z = 0$.

If two edges have the same color x, but one different y, then the value of the face is $2x+y = -x+y \neq 0$ because $x \neq y$.

We convert this to a graph theory problem. The regular icosahedron has 30 edges, 20 faces, and 12 vertices, it can be represented by a graph with 30 edges and 12 vertices.

It's **dual shape** has 30 edges, 12 faces, and 20 vertices (this is a dodecahedron), where faces in the primal are vertices in the dual, so this becomes an undirected graph of 20 vertices with weighted edges and vertices,

and we want to find all such graphs where the weight at each vertex, which is defined as the sum of all edge weights going into the vertex, is not 0 mod 3.

Then the answer is $2^{20}3^{10}$.